MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED LOW POWER SCHOTTKY TTL, AND, OR, NAND, AND NOR GATES, MONOLITHIC SILICON

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, ADVANCED, low-power Schottky TTL, positive AND, OR, NAND, AND NOR logic gating microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quadruple 2-input NAND gate, buffer output
02	Quadruple 2-input NOR gate, buffer output
03	Quadruple 2-input NAND gate, open collector, buffer output
04	Quadruple 2-input AND gate, buffer output
05	Triple 3-input NAND gate, buffer output
06	Triple 3-input AND gate, buffer output
07	Dual 4-input NAND gate, buffer output
08	Quadruple 2-input OR gate, buffer output
09	Hex 1-input inverter gate, buffer output
10	Hex 1-input inverter gate, open collector, buffer output
11	Hex 1-input noninverting gate, buffer output
12	Hex 1-input noninverting gate, open collector, buffer output

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4", flat-pack)
B	F-3 (14-lead, 3/16" x 1/4", flat-pack)
C	D-1 (14-lead, 1/4" x 3/4", dual-in-line)
D	F-2 (14-lead, 1/4" x 3/8", flat-pack)
2	C-2 (20 terminal, .350" x .350", square chip carrier package)

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to 7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to 7.0 V dc
Storage temperature range	-65°C to +150°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Maximum power dissipation (P_D) per device 1/	
Device type 01, 03	35.2 mW dc
Device type 02, 04	44 mW dc
Device type 05, 06	33 mW dc
Device type 07	17.6 mW dc
Device type 08	55 mW dc
Device type 09, 10	66 mW dc
Device type 11, 12	77 mW dc
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, B, D	70°C/W
Case C	50°C/W
Case 2	60°C/W
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Case operating temperature range (T_C)	-55°C to +125°C

2 APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

1/ Must withstand the added P_D due to short circuit test (e.g., I_O).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, I_{OH} = -1.0 mA, V_{IL} = 0.8 V	01,02,04,05,06,07,08,09,11	2.4	--	V dc
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, I_{IN} = 12 mA, V_{IH} = 2.0 V	ALL	--	0.4	V dc
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, T_C = +25°C, I_{IN} = -18 mA	ALL	--	-1.5	V dc
Low level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	ALL	0	-100	µA
High level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	ALL		20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V	ALL		100	µA
Output current 1/	I_O	V_{CC} = 5.5 V, V_O = 2.25 V	01,02,04,05,06,07,08	-15	-112	mA
			09,11	-30	-112	mA
Collector cut-off current	I_{CEX}	V_{CC} = 4.5 V, V_{IN} = 5.0 V, V_O = 5.5 V	03,10,12		100	µA
Supply current, outputs high	I_{CCH}	V_{CC} = 5.5 V	01,03		1.6	mA
			02		2.8	
			04,09,10		3	
			05		1.2	
			06		2.3	
			07		0.8	
			11,12		6	

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55^{\circ}C \leq T_C \leq +125^{\circ}C$		Device type	Limits Min	Limits Max	Unit
Supply current, outputs low	I_{CCL}	V_{CC} = 5.5 V		01,03		7.8	mA
				02		9.0	
				04,		9.3	
				05		6.0	
				06		7.0	
				07		3.9	
				08		10.6	
				09,10		12.0	
				11,12		14.0	
Propagation delay time low-to high level	t_{PLH}	V_{CC} = 5.0 V C_L = 50 pF± 10%	R_L = 500 Ω	01,02,05 07,11	2	10	ns
				04	2	11	
				08	2	9	
				06	2	12	
				09	1	9	
			R_L = 680 Ω	03	10	40	
				10,12	5	35	
Propagation delay time high-to-low level	t_{PHL}	V_{CC} = 5.0 V C_L = 50 pF± 10%	R_L = 500 Ω	01,02,05 07	2	7	ns
				04,06	3	11	
				08	3	12	
				09	1	8	
				11	1	10	
			R_L = 680 Ω	03	2	12	
				10	2	12	
				12	2	14	

1/ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.

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3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3 9,10,11	1*2,3, 9
Group A test requirements (method 5005)	1,2,3, 9,10,11	1,2,3, 9
Group B test requirements (method 5005) subgroup 5	1,2,3, 9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

c. Subgroups 3 and 4 shall be added to group C inspection for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

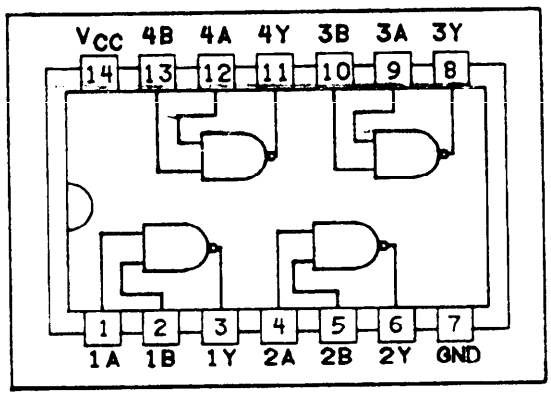
4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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Device type 01 and 03
Cases A, B, C, and D



Device type 01 and 03
Case 2

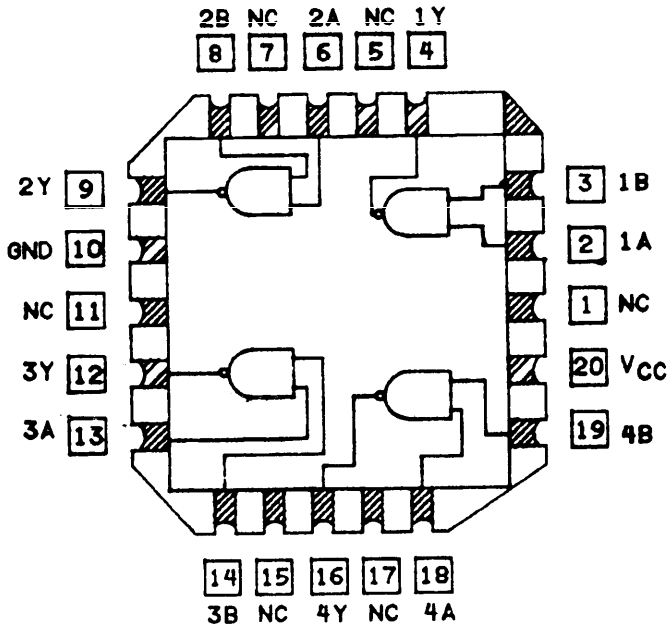
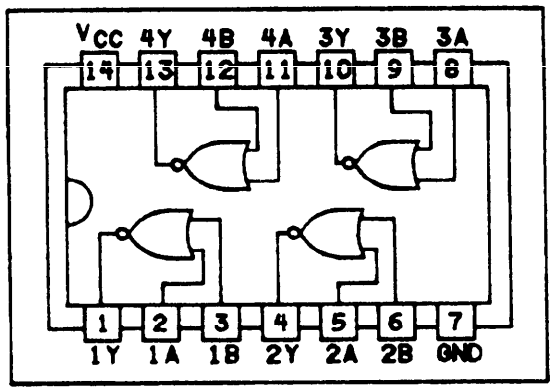


FIGURE 1. Logic diagram and terminal connections (top view).

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Device type 02
Cases A, B, C, and D



Device type 02
Case 2

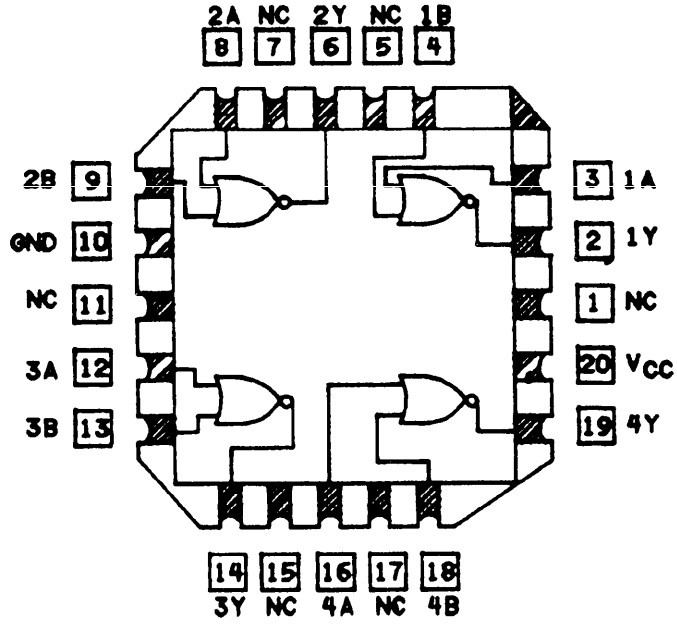
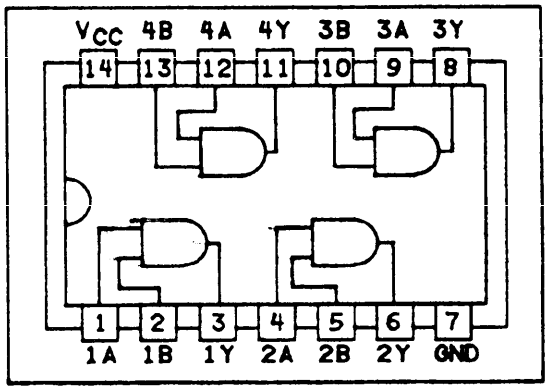


FIGURE 1. Logic diagram and terminal connections (top view).

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Device type 04
Cases A, B, C, and D



Device type 04
Case 2

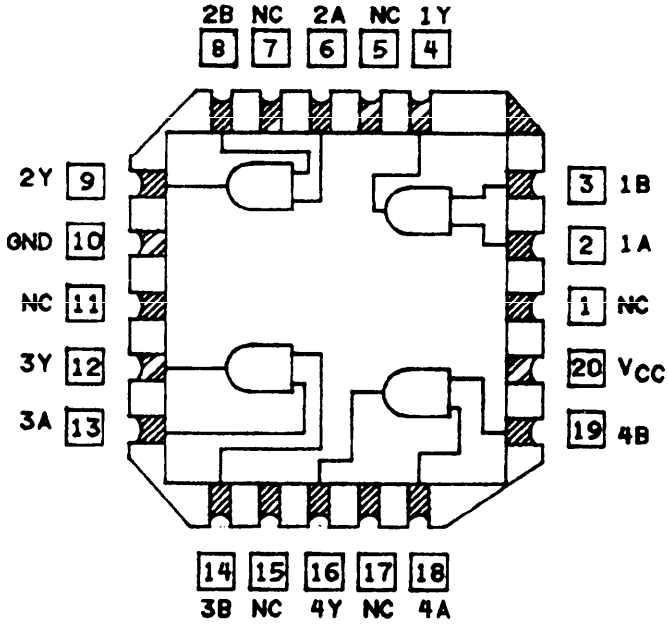
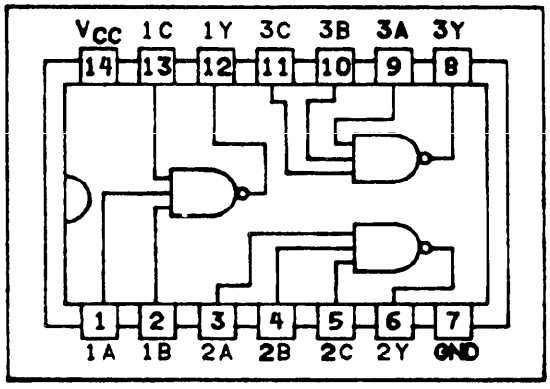


FIGURE 1. Logic diagram and terminal connections (top view).

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Device type 05
Cases A, B, C, and D



Device type 05
Case 2

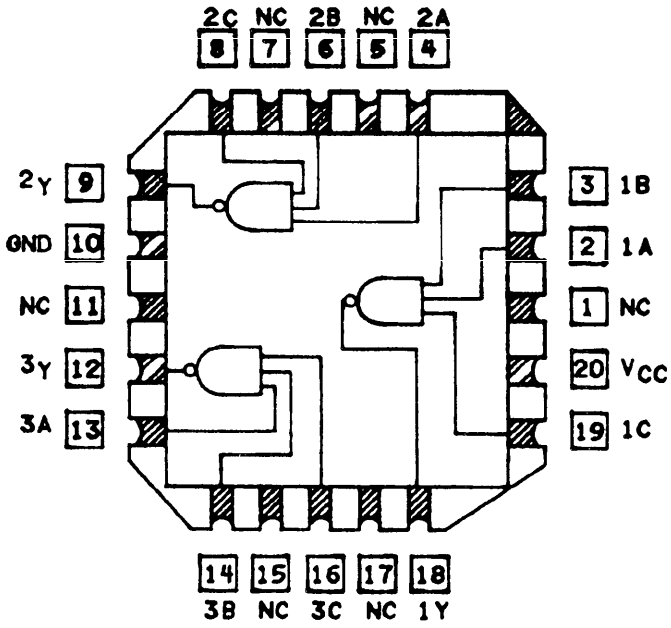
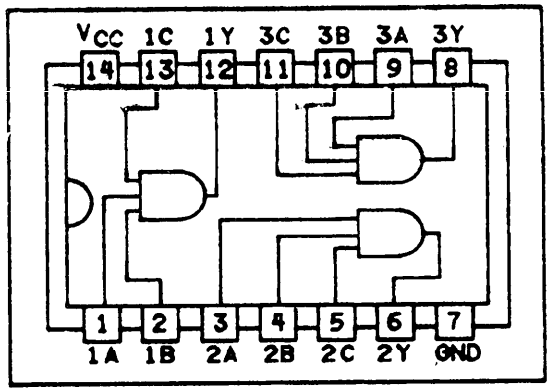


FIGURE 1. Logic diagram and terminal connections (top view).

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Device type 06
Cases A, B, C, and D



Device type 06
Case 2

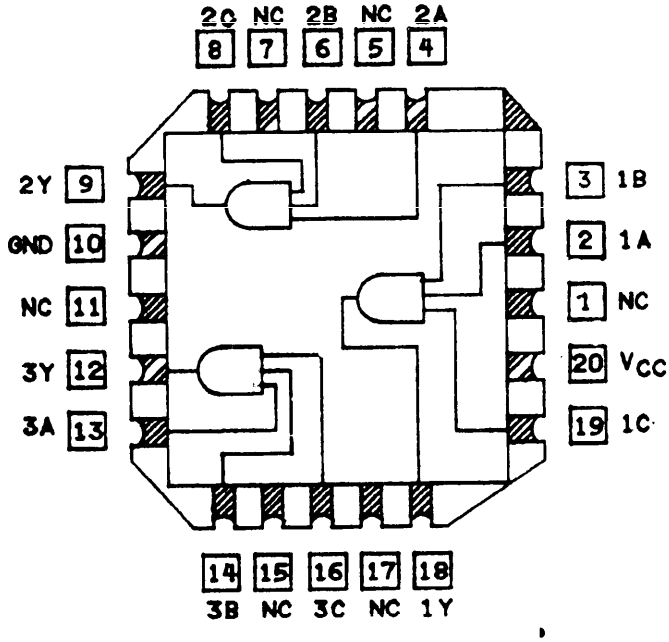
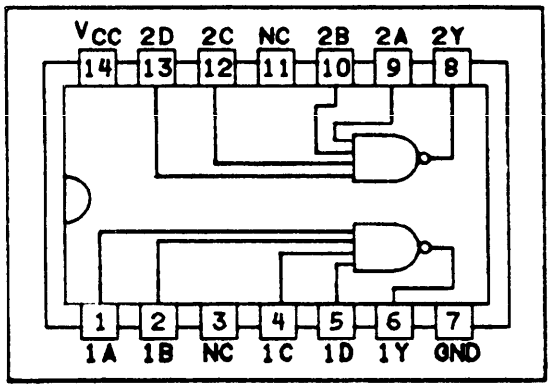


FIGURE 1. Logic diagram and terminal connections (top view)

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Device type 07
Cases A, B, C, and D



Device type 07
Case 2

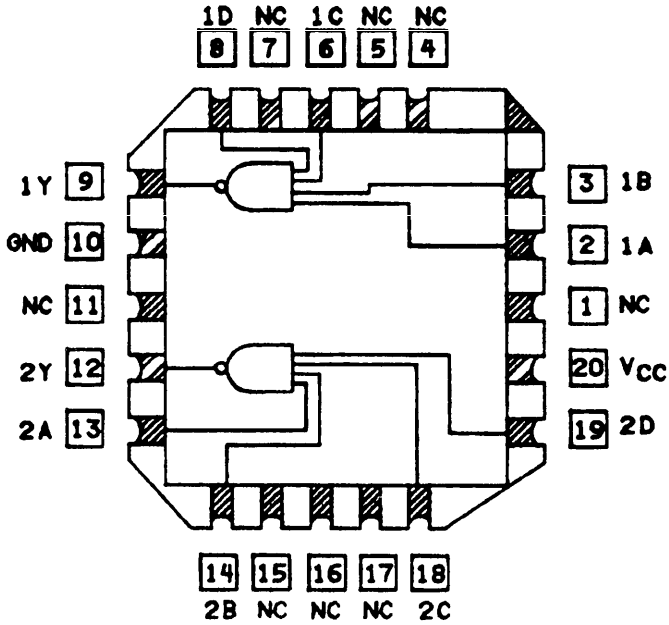
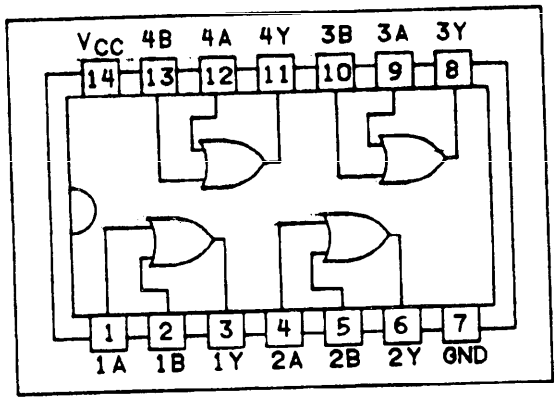


FIGURE 1. Logic diagram and terminal connections (top view).

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Device type 08
Cases A, B, C, and D



Device type 08
Case 2

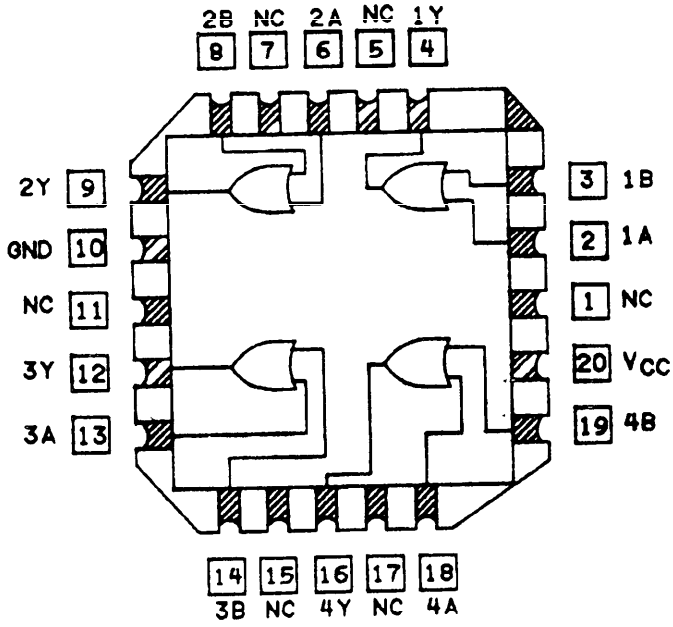
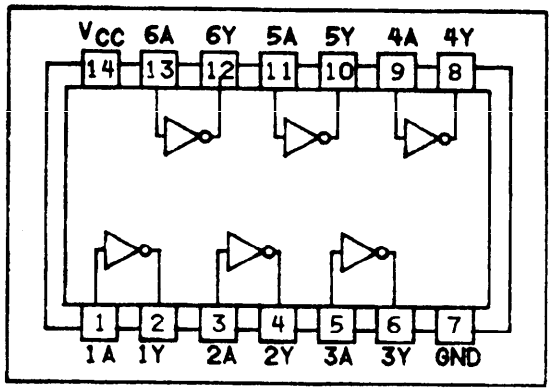


FIGURE 1. Logic diagram and terminal connections (top view).

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Device type 09 and 10
Cases A, B, C, and D



Device type 09 and 10
Case 2

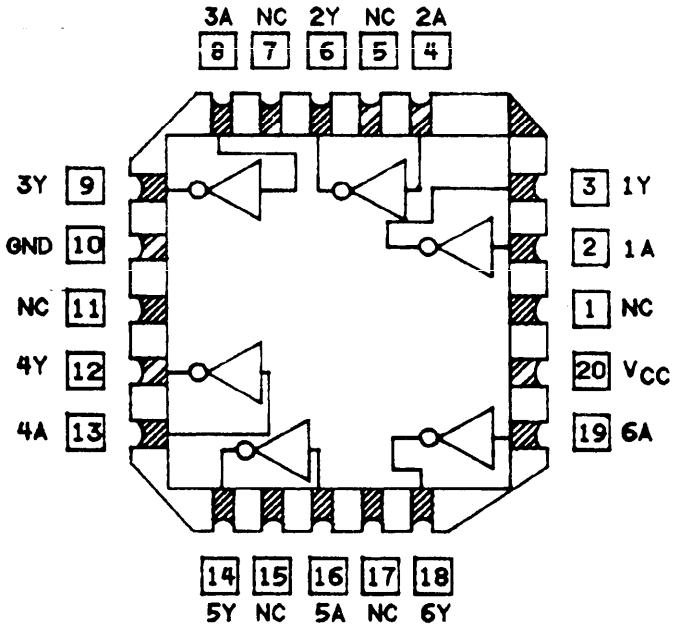
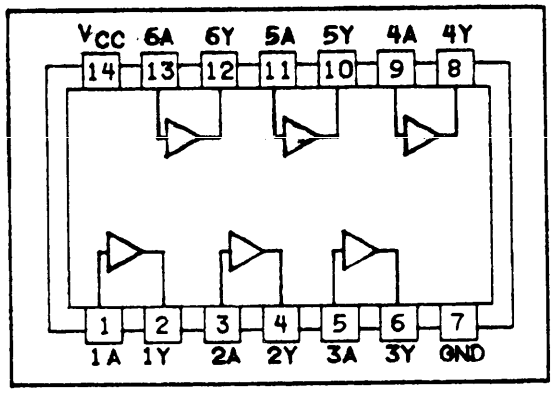


FIGURE 1. Logic diagram and terminal connections (top view).

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Device type 11 and 12
Cases A, B, C, and D



Device type 11 and 12
Case 2

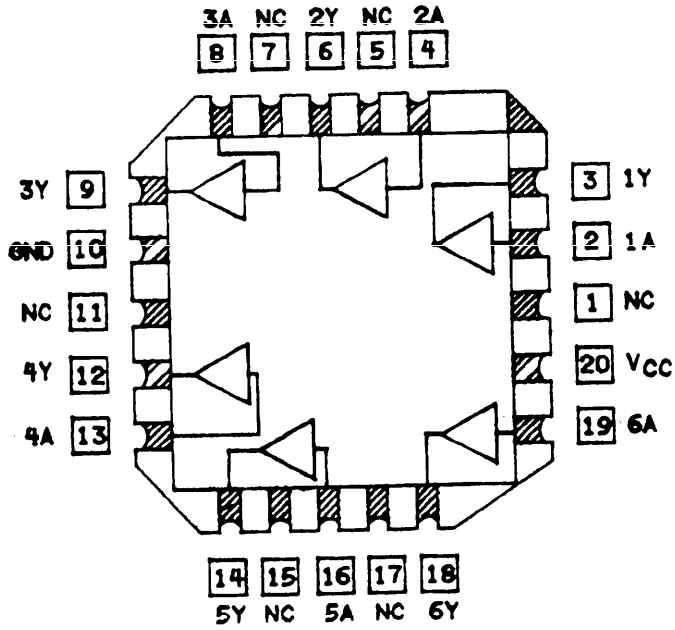


FIGURE 1. Logic diagram and terminal connections (top view).

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End-point electrical parameter limits - Continued

Device Types 01, 03

TRUTH TABLE (EACH GATE)		
INPUTS		OUTPUT
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

$Y = \overline{AB}$

Device Type 02

TRUTH TABLE (EACH GATE)		
INPUTS		OUTPUT
A	B	Y
L	L	H
H	L	L
L	H	L
H	H	L

$Y = \overline{A + B}$

Device Type 04

TRUTH TABLE (EACH GATE)		
INPUTS		OUTPUT
A	B	Y
L	L	L
H	L	L
L	H	L
H	H	H

$Y = AB$

Device Type 05

TRUTH TABLE (EACH GATE)			
INPUTS			OUTPUT
A	B	C	Y
L	L	L	H
H	L	L	H
L	H	L	H
L	L	H	H
H	H	L	H
H	L	H	H
L	H	H	H
H	H	H	L

$Y = \overline{ABC}$

Device Type 06

TRUTH TABLE (EACH GATE)			
INPUTS			OUTPUT
A	B	C	Y
L	L	L	L
H	L	L	L
L	H	L	L
L	L	H	L
H	H	L	L
H	L	H	L
L	H	H	L
H	H	H	H

$Y = ABC$

Device Type 07

TRUTH TABLE (EACH GATE)				
INPUTS				OUTPUT
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
L	L	H	L	H
L	L	L	H	H
H	H	L	L	H
H	L	H	L	H
H	L	L	H	H
L	H	H	L	H
L	H	L	H	H
L	L	H	H	H
H	L	H	H	H
H	H	L	H	H
L	H	H	H	H
H	H	H	H	L

$Y = \overline{ABCD}$

Device Type 08

TRUTH TABLE (EACH GATE)		
INPUTS		OUTPUT
A	B	Y
L	L	L
H	L	H
L	H	H
H	H	H

$Y = A + B$

Device types 09 and 10

TRUTH TABLE EACH GATE	
INPUT	OUTPUT
A	Y
L	H
H	L

$Y = \overline{A}$

Device types 11 and 12

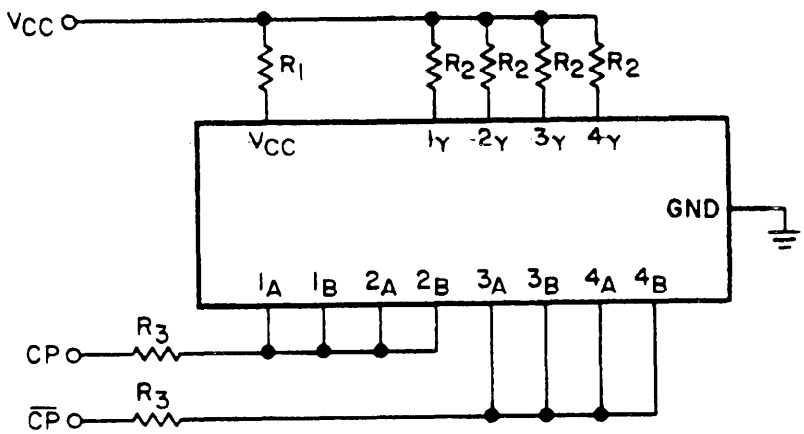
TRUTH TABLE EACH GATE	
INPUT	OUTPUT
A	Y
H	H
L	L

$Y = A$

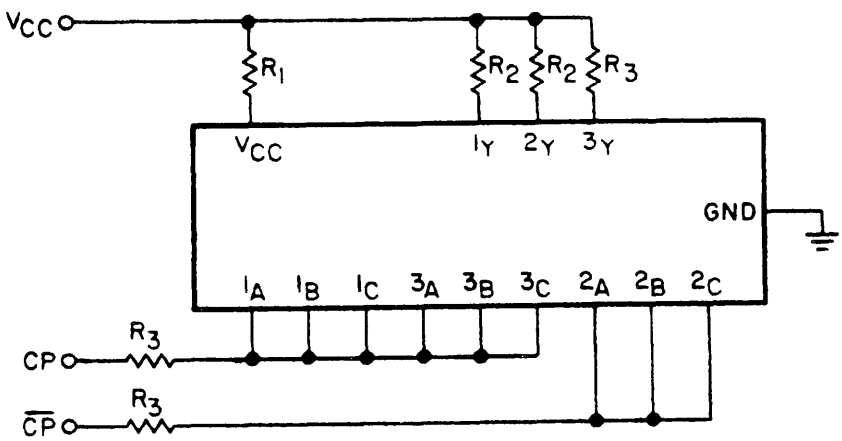
FIGURE 2. Truth tables and logic equations.

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Device Types 01, 02, 03, 04, and 08



Device types 05 and 06

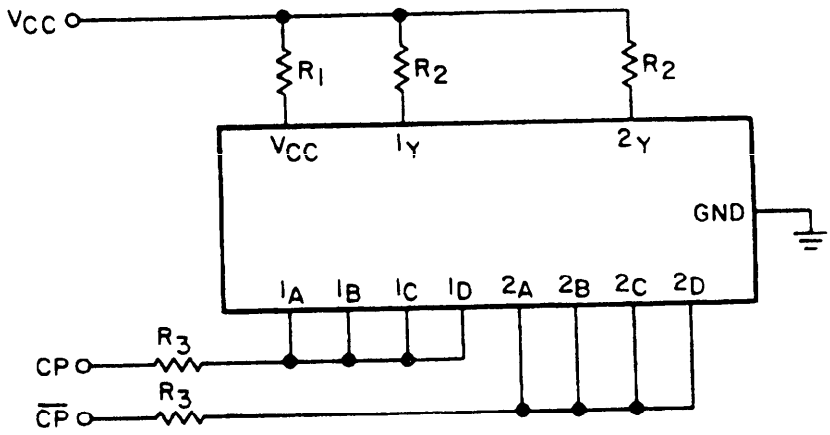


NOTES:
1. CP or $\overline{CP}$ = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R_2 = 470Ω ±5%; R_3 = 27Ω maximum.
3. R_1 and V_{CC} shall be chosen to insure 5.0 V minimum is present at device V_{CC} terminal.

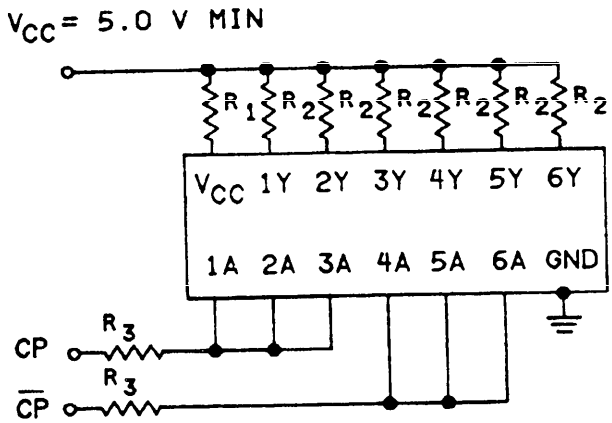
FIGURE 3. Burn-in and life test circuits.

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Device Type 07



Device types 09, 10, 11, and 12

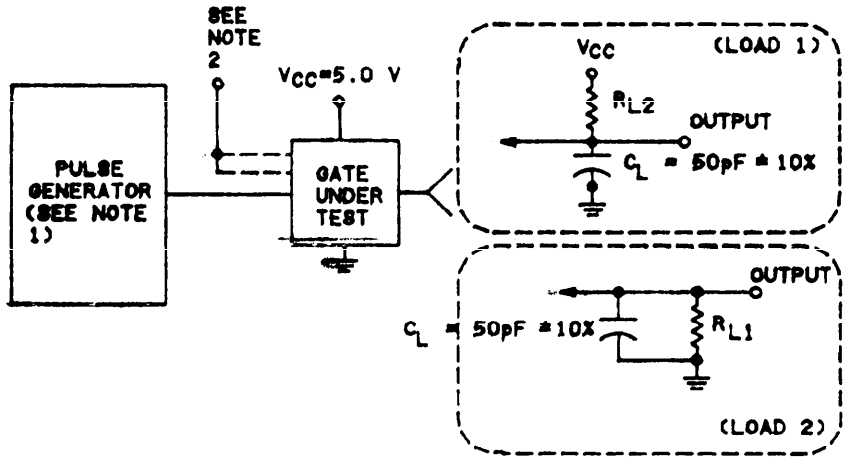


NOTES:
1. CP or $\overline{CP}$ = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R_2 = 470Ω ±5%; R_3 = 27Ω maximum.
3. R_1 and V_{CC} shall be chosen to insure 5.0 V minimum is present at device V_{CC} terminal.

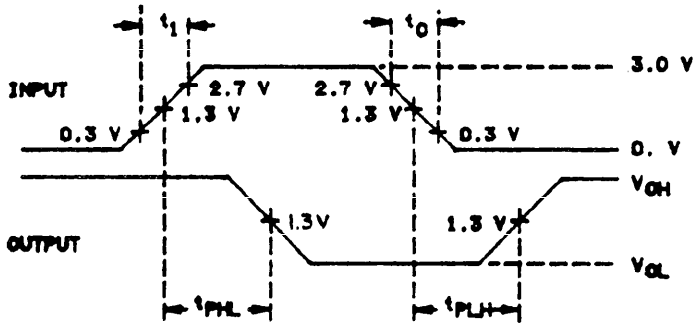
FIGURE 3. Burn-in and life test circuits - Continued.

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Device types 01, 02, 03, 05, 07, 09, and 10



TEST CIRCUIT



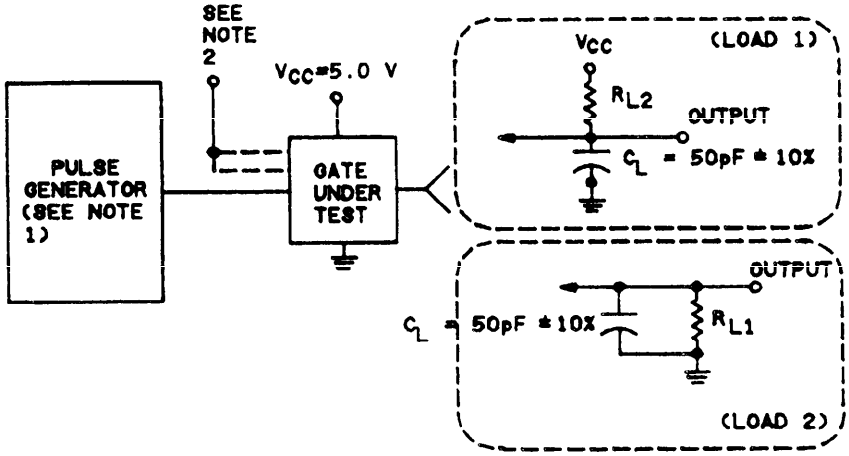
VOLTAGE WAVEFORMS

NOTES:
1. Input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; PRR $\leq$ 1 MHz; duty cycle = 50%.
2. Inputs not under test, for device types 01, 03, 05, and 07 are 2.7 V; for device type 02 is ground.
3. C_L = 50 pF ±10%, includes jig and probe capacitance.
4. R_{L1} = 499Ω ±1% for device types 01, 02, 05, 07 and 09; R_{L2} = 680Ω ±5% for device types 03 and 10.
5. Voltage measurements are made with respect to ground terminal.

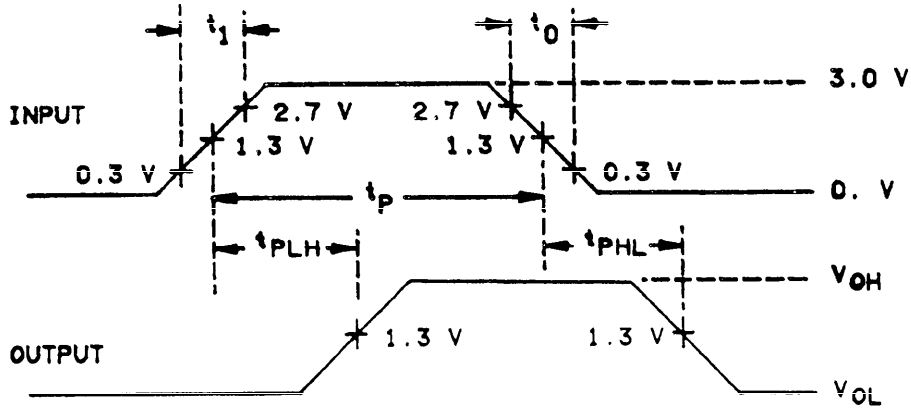
FIGURE 4. Switching time test circuit and waveforms for device types 01, 02, 03, 05, 07, 09 and 10.

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Device types 04, 06, 08, 11, and 12



TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. Input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; PRR $\leq$ 1 MHz; duty cycle = 50%.
2. Inputs not under test, for device types 04, and 06 are 2.7 V; for device type 08 is ground.
3. C_L = 50 pF ±10%, includes jig and probe capacitance.
4. R_{L1} = 499Ω ±1% for device types 04, 06, 08, and 11; R_{L2} = 680Ω ±5% for device types 12.
5. Voltage measurements are made with respect to ground terminal.

FIGURE 4. Switching time test circuit and waveforms for device types 04, 06, 08, 11, and 12.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high $\geq$ 2.0 V or low $\leq$ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	2.0 V	-1.0 mA				GND							4.5 V	1Y	2.4		V dc
		"	2	2.0 V	0.8 V	-1.0 mA				"							"	1Y	"		"
		"	3				0.8 V	2.0 V	-1.0 mA	"							"	2Y	"		"
		"	4				2.0 V	0.8 V	-1.0 mA	"							"	2Y	"		"
		"	5							"	-1.0 mA	0.8 V	2.0 V				"	3Y	"		"
		"	6							"	-1.0 mA	2.0 V	0.8 V				"	3Y	"		"
		"	7							"				-1.0 mA	0.8 V	2.0 V	"	4Y	"		"
		"	8							"				-1.0 mA	2.0 V	0.8 V	"	4Y	"		"
	V_{OL}	3007	9	2.0 V	2.0 V	12 mA				"							"	1Y		0.4	"
		"	10				2.0 V	2.0 V	12 mA	"							"	2Y		"	"
		"	11							"	12 mA	2.0 V	2.0 V				"	3Y		"	"
		"	12							"				12 mA	2.0 V	2.0 V	"	4Y		"	"
	V_{IC}		13	-18 mA						"							"	1A		-1.5	"
			14		-18 mA					"							"	1B		"	"
			15				-18 mA			"							"	2A		"	"
			16					-18 mA		"							"	2B		"	"
			17							"		-18 mA					"	3A		"	"
			18							"			-18 mA				"	3B		"	"
			19							"					-18 mA		"	4A		"	"
			20							"						-18 mA	"	4B		"	"
	I_{IL}	3009	21	0.4 V						"							5.5 V	1A	2/	2/	μA
		"	22		0.4 V					"							"	1B	"	"	"
		"	23				0.4 V			"							"	2A	"	"	"
		"	24					0.4 V		"							"	2B	"	"	"
		"	25							"		0.4 V					"	3A	"	"	"
		"	26							"			0.4 V				"	3B	"	"	"
		"	27							"					0.4 V		"	4A	"	"	"
		"	28							"						0.4 V	"	4B	"	"	"
	I_{IH1}	3010	29	2.7 V						"							"	1A		20	"
		"	30		2.7 V					"							"	1B		"	"
		"	31				2.7 V			"							"	2A		"	"
		"	32					2.7 V		"							"	2B		"	"
		"	33							"		2.7 V					"	3A		"	"
		"	34							"			2.7 V				"	3B		"	"
		"	35							"					2.7 V		"	4A		"	"
		"	36							"						2.7 V	"	4B		"	"
	I_{IH2}	"	37	7.0 V						"							"	1A		100	"
		"	38		7.0 V					"							"	1B		"	"
		"	39				7.0 V			"							"	2A		"	"
		"	40					7.0 V		"							"	2B		"	"
		"	41							"		7.0 V					"	3A		"	"
		"	42							"			7.0 V				"	3B		"	"
		"	43							"					7.0 V		"	4A		"	"
		"	44							"						7.0 V	"	4B		"	"
	I_O	3011 3/ 4/	45	GND	GND	2.25 V				"							"	1Y	-30	-112	mA
		"	46				GND	GND	2.25 V	"	2.25 V	GND	GND				"	2Y	"	"	"
		"	47							"				2.25 V	GND	GND	"	3Y	"	"	"
		"	48							"							"	4Y	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 1A	2 3 1B	3 4 1Y	4 6 2A	5 8 2B	6 9 2Y	7 10 GND	8 12 3Y	9 13 3A	10 14 3B	11 16 4Y	12 18 4A	13 19 4B	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{CCH}	3005	49	GND	GND		GND	GND		GND		GND	GND		GND	GND	5.5 V	V_{CC}		1.6	mA
	I_{CCL}	3005	50	4.5 V	4.5 V		4.5 V	4.5 V		GND		4.5 V	4.5 V		4.5 V	4.5 V	5.5 V	V_{CC}		7.8	mA
2	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PLH}	3003 Fig 4	51	IN	2.7 V	OUT				GND							5.0 V	1A to 1Y	2	8	ns
		"	52	2.7 V	IN	OUT				"							"	1B to 1Y	"	"	"
		"	53				IN	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	54				2.7 V	IN	OUT	"							"	2B to 2Y	"	"	"
		"	55							"	OUT	IN	2.7 V				"	3A to 3Y	"	"	"
		"	56							"	OUT	2.7 V	IN				"	3B to 3Y	"	"	"
		"	57							"				OUT	IN	2.7 V	"	4A to 4Y	"	"	"
		"	58							"				OUT	2.7 V	IN	"	4B to 4Y	"	"	"
	t_{PHL}	"	59	IN	2.7 V	OUT				"							"	1A to 1Y	3	8	"
		"	60	2.7 V	IN	OUT				"							"	1B to 1Y	"	"	"
		"	61				IN	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	62				2.7 V	IN	OUT	"							"	2B to 2Y	"	"	"
		"	63							"	OUT	IN	2.7 V				"	3A to 3Y	"	"	"
		"	64							"	OUT	2.7 V	IN				"	3B to 3Y	"	"	"
		"	65							"				OUT	IN	2.7 V	"	4A to 4Y	"	"	"
		"	66							"				OUT	2.7 V	IN	"	4B to 4Y	"	"	"
10 T_C = +125°C	t_{PLH} t_{PHL}	Same tests, and terminal conditions as subgroup 9 except T_C = +125°C.																	2 3	10 10	" "
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																				

1/ Pins not referenced are N/C.

2/ I_{IL} limits in μA as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-100	0/-100	0/-100

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

4/ I_O min/max limit for circuits B and C shall be: -30/-110 mA.

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	1Y	1A	1B	2Y	2A	2B	GND	3A	3B	3Y	4A	4B	4Y	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	-1.0 mA	0.8 V	0.8 V	-1.0 mA	0.8 V	0.8 V	GND	0.8 V	0.8 V	-1.0 mA	0.8 V	0.8 V	-1.0 mA	4.5 V	1Y	2.4		V dc
		"	2							"							"	2Y	"			"
		"	3							"							"	3Y	"			"
		"	4							"							"	4Y	"			"
	V_{OL}	3007	5	12 mA	2.0 V	0.8 V												1Y		0.4	
		"	6	12 mA	0.8 V	2.0 V												1Y		"	
		"	7				12 mA	2.0 V	0.8 V									2Y		"	
		"	8				12 mA	0.8 V	2.0 V									2Y		"	
		"	9								2.0 V	0.8 V	12 mA					3Y		"	
		"	10								0.8 V	2.0 V	12 mA					3Y		"	
		"	11											2.0 V	0.8 V	12 mA		4Y		"	
		"	12											0.8 V	2.0 V	12 mA		4Y		"	
	V_{IC}		13		-18 mA	-18 mA		-18 mA	-18 mA		-18 mA	-18 mA		-18 mA	-18 mA			1A		-1.5	
			14															1B		"	
			15															2A		"	
			16															2B		"	
			17															3A		"	
			18															3B		"	
			19															4A		"	
			20															4B		"	
	I_{IL}	3009	21		0.4 V	0.4 V		0.4 V	0.4 V		0.4 V	0.4 V		0.4 V	0.4 V		5.5 V	1A	2/	2/	uA
		"	22														"	1B	"	"	"
		"	23														"	2A	"	"	"
		"	24														"	2B	"	"	"
		"	25														"	3A	"	"	"
		"	26														"	3B	"	"	"
		"	27														"	4A	"	"	"
		"	28														"	4B	"	"	"
	I_{IH1}	3010	29		2.7 V	2.7 V		2.7 V	2.7 V		2.7 V	2.7 V		2.7 V	2.7 V			1A		20	
		"	30														"	1B		"	
		"	31														"	2A		"	
		"	32														"	2B		"	
		"	33														"	3A		"	
		"	34														"	3B		"	
		"	35														"	4A		"	
		"	36														"	4B		"	
	I_{IH2}	"	37		7.0 V	7.0 V		7.0 V	7.0 V		7.0 V	7.0 V		7.0 V	7.0 V			1A		100	
		"	38														"	1B		"	
		"	39														"	2A		"	
		"	40														"	2B		"	
		"	41														"	3A		"	
		"	42														"	3B		"	
		"	43														"	4A		"	
		"	44														"	4B		"	
	I_O	3011 3/ 4/	45	2.25 V	GND	GND	2.25 V	GND	GND		GND	GND	2.25 V	GND	GND	2.25 V		1Y	-30	-112	mA
		"	46														"	2Y	"	"	"
		"	47														"	3Y	"	"	"
		"	48														"	4Y	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 - Continued.

Terminal conditions (pins not designated may be high > 2.0 V or low < 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 1Y	2 3 1A	3 4 1B	4 6 2Y	5 8 2A	6 9 2B	7 10 GND	8 12 3A	9 13 3B	10 14 3Y	11 16 4A	12 18 4B	13 19 4Y	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{CCH}	3005	49		GND	GND		GND	GND	GND	GND	GND		GND	GND		5.5 V	V_{CC}		2.8	mA
	I_{CCL}	3005	50		4.5 V	4.5 V		4.5 V	4.5 V	GND	4.5 V	4.5 V		4.5 V	4.5 V		5.5 V	V_{CC}		9.0	mA
2	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PLH}	3003 Fig 4	51	OUT	IN	GND				GND							5.0 V	1A to 1Y	2	8	ns
			52	OUT	GND	IN	OUT	IN	GND	"	IN	GND	OUT				"	1B to 1Y	"	"	"
			53				OUT	GND	IN	"	GND	IN	OUT				"	2A to 2Y	"	"	"
			54							"				IN	GND	OUT	"	2B to 2Y	"	"	"
			55							"				GND	IN	OUT	"	3A to 3Y	"	"	"
			56							"							"	3B to 3Y	"	"	"
			57							"							"	4A to 4Y	"	"	"
			58							"							"	4B to 4Y	"	"	"
	t_{PHL}	"	59	OUT	IN	GND				GND							"	1A to 1Y	3	8	"
		"	60	OUT	GND	IN	OUT	IN	GND	"	IN	GND	OUT				"	1B to 1Y	"	"	"
		"	61				OUT	GND	IN	"	GND	IN	OUT				"	2A to 2Y	"	"	"
		"	62							"				IN	GND	OUT	"	2B to 2Y	"	"	"
		"	63							"				GND	IN	OUT	"	3A to 3Y	"	"	"
		"	64							"							"	3B to 3Y	"	"	"
		"	65							"							"	4A to 4Y	"	"	"
		"	66							"							"	4B to 4Y	"	"	"
10 T_C = +125°C	t_{PLH} t_{PHL}	Same tests, and terminal conditions as subgroup 9 except T_C = +125°C.																	2 3	10 10	" "
11	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																				

1/ Pins not referenced are N/C.

2/ I_{IL} limits in μA as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-100	0/-100	0/-100

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

4/ I_O min/max limit for circuits B and C shall be: -30/-110 mA.

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}				
1 T_C = +25°C	V_{OL}	3006	1	2.0 V	2.0 V	12 mA				GND							4.5 V	1Y		0.4	V dc
		"	2				2.0 V	2.0 V	12 mA	"							"	2Y		"	"
		"	3							"	12 mA	2.0 V	2.0 V				"	3Y		"	"
		"	4							"				12 mA	2.0 V	2.0 V	"	4Y		"	"
	V_{IC}		5	-18 mA						"							"	1A		-1.5	"
			6		-18 mA					"							"	1B		"	"
			7				-18 mA			"							"	2A		"	"
			8					-18 mA		"							"	2B		"	"
			9							"		-18 mA					"	3A		"	"
			10							"			-18 mA				"	3B		"	"
			11							"					-18 mA		"	4A		"	"
			12							"						-18 mA	"	4B		"	"
	I_{IL}	3009	13	0.4 V						"							5.5 V	1A	2/	2/	uA
		"	14		0.4 V					"							"	1B	"	"	"
		"	15				0.4 V			"							"	2A	"	"	"
		"	16					0.4 V		"							"	2B	"	"	"
		"	17							"		0.4 V					"	3A	"	"	"
		"	18							"			0.4 V				"	3B	"	"	"
		"	19							"					0.4 V		"	4A	"	"	"
		"	20							"						0.4 V	"	4B	"	"	"
	I_{IH1}	3010	21	2.7 V						"							"	1A		20	"
		"	22		2.7 V					"							"	1B		"	"
		"	23				2.7 V			"							"	2A		"	"
		"	24					2.7 V		"							"	2B		"	"
		"	25							"		2.7 V					"	3A		"	"
		"	26							"			2.7 V				"	3B		"	"
		"	27							"					2.7 V		"	4A		"	"
		"	28							"						2.7 V	"	4B		"	"
	I_{IH2}	"	29	7.0 V						"							"	1A		100	"
		"	30		7.0 V					"							"	1B		"	"
		"	31				7.0 V			"							"	2A		"	"
		"	32					7.0 V		"							"	2B		"	"
		"	33							"		7.0 V					"	3A		"	"
		"	34							"			7.0 V				"	3B		"	"
		"	35							"					7.0 V		"	4A		"	"
		"	36							"						7.0 V	"	4B		"	"
	I_{CCX}	"	37	0.8 V	5.0 V	5.5 V				"							4.5 V	1Y		100	"
		"	38	5.0 V	0.8 V	5.5 V				"							"	1Y		"	"
		"	39				0.8 V	5.0 V	5.5 V	"							"	2Y		"	"
		"	40				5.0 V	0.8 V	5.5 V	"							"	2Y		"	"
		"	41							"	5.5 V	0.8 V	5.0 V				"	3Y		"	"
		"	42							"	5.5 V	5.0 V	0.8 V				"	3Y		"	"
		"	43							"				5.5 V	0.8 V	5.0 V	"	4Y		"	"
		"	44							"				5.5 V	5.0 V	0.8 V	"	4Y		"	"
	I_{CCH}	3005	45	GND	GND		GND	GND		"		GND	GND		GND	GND	5.5 V	V_{CC}		1.6	mA
	I_{CCL}	3005	46	4.5 V	4.5 V		4.5 V	4.5 V		"		4.5 V	4.5 V		4.5 V	4.5 V	5.5 V	V_{CC}		7.8	mA

See footnotes at end of table.

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TABLE III. Group A inspection for device type 03 - Continued.

Terminal conditions (pins not designated may be high > 2.0 V or low < 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 1A	2 2 3B	3 4 1Y	4 6 2A	5 8 2B	6 9 2Y	7 10 GND	8 12 3Y	9 13 3A	10 14 3B	11 16 4Y	12 18 4A	13 19 4B	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	t_{PLH}	3003 Fig. 4	47	IN 2.7 V	2.7 V IN	OUT OUT				GND							5.0 V	1A to 1Y	10	33	ns
		"	48							"							"	1B to 1Y	"	"	"
		"	49				IN 2.7 V	2.7 V IN	OUT OUT	"							"	2A to 2Y	"	"	"
		"	50							"							"	2B to 2Y	"	"	"
		"	51							"	OUT OUT	IN 2.7 V	2.7 V IN				"	3A to 3Y	"	"	"
		"	52							"							"	3B to 3Y	"	"	"
		"	53							"				OUT OUT	IN 2.7 V	2.7 V IN	"	4A to 4Y	"	"	"
		"	54							"							"	4B to 4Y	"	"	"
	t_{PHL}	"	55	IN 2.7 V	2.7 V IN	OUT OUT				"							"	1A to 1Y	7	18	"
		"	56							"							"	1B to 1Y	"	"	"
		"	57				IN 2.7 V	2.7 V IN	OUT OUT	"							"	2A to 2Y	"	"	"
		"	58							"							"	2B to 2Y	"	"	"
		"	59							"	OUT OUT	IN 2.7 V	2.7 V IN				"	3A to 3Y	"	"	"
		"	60							"							"	3B to 3Y	"	"	"
		"	61							"				OUT OUT	IN 2.7 V	2.7 V IN	"	4A to 4Y	"	"	"
		"	62							"							"	4B to 4Y	"	"	"
10 T_C = +125°C	t_{PLH} t_{PHL}	Same tests, and terminal conditions as subgroup 9 except T_C = +125°C.																	10 7	40 21	" "
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																				

1/ Pins not referenced are N/C.

2/ I_{IL} limits in µA as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-100	0/-100	0/-100

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TABLE III. Group A inspection for device type 04.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	2.0 V	-1.0 mA				GND							4.5 V	1Y	2.4		V dc
		"	2				2.0 V	2.0 V	-1.0 mA	"							"	2Y	"		"
		"	3							"	-1.0 mA	2.0 V	2.0 V				"	3Y	"		"
		"	4							"				-1.0 mA	2.0 V	2.0 V	"	4Y	"		"
	V_{OL}	3007	5	0.8 V	5.5 V	12 mA				"							"	1Y		0.4	"
		"	6	5.5 V	0.8 V	12 mA				"							"	1Y		"	"
		"	7				0.8 V	5.5 V	12 mA	"							"	2Y		"	"
		"	8				5.5 V	0.8 V	12 mA	"							"	2Y		"	"
		"	9							"	12 mA	0.8 V	5.5 V				"	3Y		"	"
		"	10							"	12 mA	5.5 V	0.8 V				"	3Y		"	"
		"	11							"				12 mA	0.8 V	5.5 V	"	4Y		"	"
		"	12							"				12 mA	5.5 V	0.8 V	"	4Y		"	"
	V_{IC}		13	-18 mA						"							"	1A		-1.5	"
			14		-18 mA					"							"	1B		"	"
			15				-18 mA			"							"	2A		"	"
			16					-18 mA		"							"	2B		"	"
			17							"		-18 mA					"	3A		"	"
			18							"			-18 mA				"	3B		"	"
			19							"					-18 mA		"	4A		"	"
			20							"						-18 mA	"	4B		"	"
	I_{IL}	3009	21	0.4 V						"							5.5 V	1A	2/	2/	μA
		"	22		0.4 V					"							"	1B	"	"	"
		"	23				0.4 V			"							"	2A	"	"	"
		"	24					0.4 V		"							"	2B	"	"	"
		"	25							"		0.4 V					"	3A	"	"	"
		"	26							"			0.4 V				"	3B	"	"	"
		"	27							"					0.4 V		"	4A	"	"	"
		"	28							"						0.4 V	"	4B	"	"	"
	I_{IH1}	3010	29	2.7 V						"							"	1A		20	μA
		"	30		2.7 V					"							"	1B		"	"
		"	31				2.7 V			"							"	2A		"	"
		"	32					2.7 V		"							"	2B		"	"
		"	33							"		2.7 V					"	3A		"	"
		"	34							"			2.7 V				"	3B		"	"
		"	35							"					2.7 V		"	4A		"	"
		"	36							"						2.7 V	"	4B		"	"
	I_{IH2}	"	37	7.0 V						"							"	1A		100	μA
		"	38		7.0 V					"							"	1B		"	"
		"	39				7.0 V			"							"	2A		"	"
		"	40					7.0 V		"							"	2B		"	"
		"	41							"		7.0 V					"	3A		"	"
		"	42							"			7.0 V				"	3B		"	"
		"	43							"					7.0 V		"	4A		"	"
		"	44							"						7.0 V	"	4B		"	"
	I_O	3011	45	5.5 V	5.5 V	2.25 V				"							"	1Y	-30	-112	mA
		3/ 4/ "	46				5.5 V	5.5 V	2.25 V	"							"	2Y	"	"	"
		"	47							"	2.25 V	5.5 V	5.5 V				"	3Y	"	"	"
		"	48							"				2.25 V	5.5 V	5.5 V	"	4Y	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 04 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 1A	2 3 1B	3 4 1Y	4 6 2A	5 8 2B	6 9 2Y	7 10 GND	8 12 3Y	9 13 3A	10 14 3B	11 16 4Y	12 18 4A	13 19 4B	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{CCH}	3005	49	4.5 V	4.5 V		4.5 V	4.5 V		GND		4.5 V	4.5 V		4.5 V	4.5 V	5.5 V	V_{CC}		3.0	mA
	I_{CCL}	3005	50	GND	GND		GND	GND		GND		GND	GND		GND	GND	5.5 V	V_{CC}		9.3	mA
2	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PLH}	3003 Fig 4	51	IN	2.7 V IN	OUT				GND							5.0 V	1A to 1Y	2	9	ns
		"	52	2.7 V		OUT				"							"	1B to 1Y	"	"	"
		"	53				IN	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	54				2.7 V	IN	OUT	"							"	2B to 2Y	"	"	"
		"	55							"	OUT	IN	2.7 V				"	3A to 3Y	"	"	"
		"	56							"	OUT	2.7 V	IN				"	3B to 3Y	"	"	"
		"	57							"				OUT	IN	2.7 V	"	4A to 4Y	"	"	"
		"	58							"				OUT	2.7 V	IN	"	4B to 4Y	"	"	"
	t_{PHL}	"	59	IN	2.7 V IN	OUT				"							"	1A to 1Y	3	9	"
		"	60	2.7 V		OUT				"							"	1B to 1Y	"	"	"
		"	61				IN	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	62				2.7 V	IN	OUT	"							"	2B to 2Y	"	"	"
		"	63							"	OUT	IN	2.7 V				"	3A to 3Y	"	"	"
		"	64							"	OUT	2.7 V	IN				"	3B to 3Y	"	"	"
		"	65							"				OUT	IN	2.7 V	"	4A to 4Y	"	"	"
		"	66							"				OUT	2.7 V	IN	"	4B to 4Y	"	"	"
10 T_C = +125°C	t_{PLH} t_{PHL}	Same tests, and terminal conditions as subgroup 9 except T_C = +125°C.																	2 3	11 11	" "
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																				

1/ Pins not referenced are N/C.

2/ I_{IL} limits in μA as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-100	0/-100	0/-100

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

4/ I_O min/max limit for circuits B and C shall be: -30/-110 mA.

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TABLE III. Group A inspection for device type 05.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 1A	2 3 1B	3 4 2A	4 6 2B	5 8 2C	6 9 2Y	7 10 GND	8 12 3Y	9 13 3A	10 14 3B	11 16 3C	12 18 1Y	13 19 1C	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	2.0 V					GND					-1.0 mA	2.0 V	4.5 V	1Y	2.4		V dc
		"	2	2.0 V	0.8 V					"					-1.0 mA	2.0 V	"	1Y	"		"
		"	3							"					-1.0 mA	0.8 V	"	1Y	"		"
		"	4			0.8 V	2.0 V	2.0 V	-1.0 mA	"							"	2Y	"		"
		"	5			2.0 V	0.8 V	2.0 V	-1.0 mA	"							"	2Y	"		"
		"	6			2.0 V	2.0 V	0.8 V	-1.0 mA	"							"	2Y	"		"
		"	7							"	-1.0 mA	0.8 V	2.0 V	2.0 V			"	3Y	"		"
		"	8							"	-1.0 mA	2.0 V	0.8 V	2.0 V			"	3Y	"		"
		"	9							"	-1.0 mA	2.0 V	2.0 V	0.8 V			"	3Y	"		"
	V_{OL}	3007	10	2.0 V	2.0 V					"					12 mA	2.0 V	"	1Y		0.4	"
		"	11			2.0 V	2.0 V	2.0 V	12 mA	"							"	2Y		"	"
		"	12							"	12 mA	2.0 V	2.0 V	2.0 V			"	3Y		"	"
	V_{IC}		13	-18 mA						"							"	1A		-1.5	"
			14		-18 mA					"							"	1B		"	"
			15							"						-18 mA	"	1C		"	"
			16			-18 mA				"							"	2A		"	"
			17				-18 mA			"							"	2B		"	"
			18					-18 mA		"							"	2C		"	"
			19							"		-18 mA					"	3A		"	"
			20							"			-18 mA				"	3B		"	"
			21							"				-18 mA			"	3C		"	"
	I_{IL}	3009	22	0.4 V						"							5.5 V	1A	2/	2/	μA
		"	23		0.4 V					"							"	1B	"	"	"
		"	24							"						0.4 V	"	1C	"	"	"
		"	25			0.4 V				"							"	2A	"	"	"
		"	26				0.4 V			"							"	2B	"	"	"
		"	27					0.4 V		"							"	2C	"	"	"
		"	28							"		0.4 V					"	3A	"	"	"
		"	29							"			0.4 V				"	3B	"	"	"
		"	30							"				0.4 V			"	3C	"	"	"
	I_{IH1}	3010	31	2.7 V						"							"	1A		20	"
		"	32		2.7 V					"							"	1B		"	"
		"	33							"						2.7 V	"	1C		"	"
		"	34			2.7 V				"							"	2A		"	"
		"	35				2.7 V			"							"	2B		"	"
		"	36					2.7 V		"							"	2C		"	"
		"	37							"		2.7 V					"	3A		"	"
		"	38							"			2.7 V				"	3B		"	"
		"	39							"				2.7 V			"	3C		"	"
	I_{IH2}	"	40	7.0 V						"							"	1A		100	"
		"	41		7.0 V					"							"	1B		"	"
		"	42							"						7.0 V	"	1C		"	"
		"	43			7.0 V				"							"	2A		"	"
		"	44				7.0 V			"							"	2B		"	"
		"	45					7.0 V		"							"	2C		"	"
		"	46							"		7.0 V					"	3A		"	"
		"	47							"			7.0 V				"	3B		"	"
		"	48							"				7.0 V			"	3C		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2 Test no.	1 / 2 / 1A	2 / 3 / 1B	3 / 4 / 2A	4 / 6 / 2B	5 / 8 / 2C	6 / 9 / 2Y	7 / 10 / GND	8 / 12 / 3Y	9 / 13 / 3A	10 / 14 / 3B	11 / 16 / 3C	12 / 18 / 1Y	13 / 19 / 1C	14 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_O	3011 3/ 4/	49	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	2.25 V	GND	2.25 V	4.5 V	4.5 V	4.5 V	2.25 V	4.5 V	5.5 V	1Y	-30	-112	mA
			50	"	"	"	"	"	"	"	"	"	"	"	"	"	"	2Y	"	"	"
			51	"	"	"	"	"	"	"	"	"	"	"	"	"	"	3Y	"	"	"
	I_{CCH}	3005	52	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		"		4.5 V	4.5 V	4.5 V		4.5 V	"	V_{CC}		1.2	"
	I_{CCL}	3005	53	GND	GND	GND	GND	GND		"		GND	GND	GND		GND	"	V_{CC}		6.0	"
2	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PLH}	3003 Fig. 4	54	IN	2.7 V					GND					OUT	2.7 V	5.0 V	1A to 1Y	2	8	ns
			55	2.7 V	IN					"					OUT	2.7 V	"	1B to 1Y	"	"	"
			56	2.7 V	2.7 V					"					OUT	IN	"	1C to 1Y	"	"	"
			57			IN	2.7 V	2.7 V	OUT	"							"	2A to 2Y	"	"	"
			58			2.7 V	IN	2.7 V	OUT	"							"	2B to 2Y	"	"	"
			59			2.7 V	2.7 V	IN	OUT	"							"	2C to 2Y	"	"	"
			60							"	OUT	IN	2.7 V	2.7 V			"	3A to 3Y	"	"	"
			61							"	OUT	2.7 V	IN	2.7 V			"	3B to 3Y	"	"	"
			62							"	OUT	2.7 V	2.7 V	IN			"	3C to 3Y	"	"	"
	t_{PHL}	"	63	IN	2.7 V					"					OUT	2.7 V	"	1A to 1Y	3	8	"
		"	64	2.7 V	IN					"					OUT	2.7 V	"	1B to 1Y	"	"	"
		"	65	2.7 V	2.7 V					"					OUT	IN	"	1C to 1Y	"	"	"
		"	66			IN	2.7 V	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	67			2.7 V	IN	2.7 V	OUT	"							"	2B to 2Y	"	"	"
		"	68			2.7 V	2.7 V	IN	OUT	"							"	2C to 2Y	"	"	"
		"	69							"	OUT	IN	2.7 V	2.7 V			"	3A to 3Y	"	"	"
		"	70							"	OUT	2.7 V	IN	2.7 V			"	3B to 3Y	"	"	"
		"	71							"	OUT	2.7 V	2.7 V	IN			"	3C to 3Y	"	"	"
10 T_C = +125°C	t_{PLH} t_{PHL}	Same tests, and terminal conditions as subgroup 9 except T_C = +125°C.																	2 3	10 10	" "
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																				

1/ Pins not referenced are N/C.

2/ I_{IL} limits in μA as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-100	0/-100	0/-100

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

4/ I_O min/max limit for circuits B and C shall be: -30/-110 mA.

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TABLE III. Group A inspection for device type 06.

Terminal conditions (pins not designated may be high > 2.0 V or low < 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2 Test no.	1 2 1A	2 3 1B	3 4 2A	4 6 2B	5 8 2C	6 9 2Y	7 10 GND	8 12 3Y	9 13 3A	10 14 3B	11 16 3C	12 18 1Y	13 19 1C	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	2.0 V					GND					-1.0 mA	2.0 V	4.5 V	1Y	2.4		V dc
		"	2			2.0 V	2.0 V	2.0 V	-1.0 mA	"							"	2Y	"		"
		"	3							"	-1.0 mA	2.0 V	2.0 V	2.0 V			"	3Y	"		"
	V_{OL}	"	4	0.8 V	2.0 V					"					12 mA	2.0 V	"	1Y		0.4	"
		"	5	2.0 V	0.8 V					"					12 mA	2.0 V	"	1Y		"	"
		"	6	2.0 V	2.0 V					"					12 mA	0.8 V	"	1Y		"	"
		"	7			0.8 V	2.0 V	2.0 V	12 mA	"							"	2Y		"	"
		"	8			2.0 V	0.8 V	2.0 V	12 mA	"							"	2Y		"	"
		"	9			2.0 V	2.0 V	0.8 V	12 mA	"							"	2Y		"	"
		"	10							"	12 mA	0.8 V	2.0 V	2.0 V			"	3Y		"	"
		"	11							"	12 mA	2.0 V	0.8 V	2.0 V			"	3Y		"	"
		"	12							"	12 mA	2.0 V	2.0 V	0.8 V			"	3Y		"	"
	V_{IC}		13	-18 mA						"							"	1A		-1.5	"
			14		-18 mA					"							"	1B		"	"
			15							"						-18 mA	"	1C		"	"
			16			-18 mA				"							"	2A		"	"
			17				-18 mA			"							"	2B		"	"
			18					-18 mA		"							"	2C		"	"
			19							"		-18 mA					"	3A		"	"
			20							"			-18 mA				"	3B		"	"
			21							"				-18 mA			"	3C		"	"
	I_{IL}	3009	22	0.4 V						"							5.5 V	1A	2/	2/	µA
		"	23		0.4 V					"							"	1B	"	"	"
		"	24							"						0.4 V	"	1C	"	"	"
		"	25			0.4 V				"							"	2A	"	"	"
		"	26				0.4 V			"							"	2B	"	"	"
		"	27					0.4 V		"							"	2C	"	"	"
		"	28							"		0.4 V					"	3A	"	"	"
		"	29							"			0.4 V				"	3B	"	"	"
		"	30							"				0.4 V			"	3C	"	"	"
	I_{IH1}	3010	31	2.7 V						"							"	1A		20	"
		"	32		2.7 V					"							"	1B		"	"
		"	33							"						2.7 V	"	1C		"	"
		"	34			2.7 V				"							"	2A		"	"
		"	35				2.7 V			"							"	2B		"	"
		"	36					2.7 V		"							"	2C		"	"
		"	37							"		2.7 V					"	3A		"	"
		"	38							"			2.7 V				"	3B		"	"
		"	39							"				2.7 V			"	3C		"	"
	I_{IH2}	"	40	7.0 V						"							"	1A		100	"
		"	41		7.0 V					"							"	1B		"	"
		"	42							"						7.0 V	"	1C		"	"
		"	43			7.0 V				"							"	2A		"	"
		"	44				7.0 V			"							"	2B		"	"
		"	45					7.0 V		"							"	2C		"	"
		"	46							"		7.0 V					"	3A		"	"
		"	47							"			7.0 V				"	3B		"	"
		"	48							"				7.0 V			"	3C		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 06 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2 1/ Test no.	1 1A	2 1B	3 2A	4 2B	5 2C	6 2Y	7 GND	8 3Y	9 3A	10 3B	11 3C	12 1Y	13 1C	14 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_O	3011 3/ 4/	49	4.5 V	4.5 V					GND											
		"	50			4.5 V	4.5 V	4.5 V	2.25 V	"					2.25 V	4.5 V	5.5 V	1Y	-30	-112	mA
		"	51							"	2.25 V	4.5 V	4.5 V	4.5 V			"	2Y	"	"	"
																	"	3Y	"	"	"
	I_{CCH}	3005	52	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		"		4.5 V	4.5 V	4.5 V		4.5 V	"	V_{CC}		2.3	"
	I_{CCL}	3005	52	GND	GND	GND	GND	GND		"		GND	GND	GND		GND	"	V_{CC}		7.0	"
2	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PLH}	3003 Fig. 4	54	IN	2.7 V					GND					OUT	2.7 V	5.0 V	1A to 1Y	2	10	ns
		"	55	2.7 V	IN					"					OUT	2.7 V	"	1B to 1Y	"	"	"
		"	56	2.7 V	2.7 V					"					OUT	IN	"	1C to 1Y	"	"	"
		"	57			IN	2.7 V	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	58			2.7 V	IN	2.7 V	OUT	"							"	2B to 2Y	"	"	"
		"	59			2.7 V	2.7 V	IN	OUT	"							"	2C to 2Y	"	"	"
		"	60							"	OUT	IN	2.7 V	2.7 V			"	3A to 3Y	"	"	"
		"	61							"	OUT	2.7 V	IN	2.7 V			"	3B to 3Y	"	"	"
		"	62							"	OUT	2.7 V	2.7 V	IN			"	3C to 3Y	"	"	"
	t_{PHL}	"	63	IN	2.7 V					"					OUT	2.7 V	"	1A to 1Y	3	9	"
		"	64	2.7 V	IN					"					OUT	2.7 V	"	1B to 1Y	"	"	"
		"	65	2.7 V	2.7 V					"					OUT	IN	"	1C to 1Y	"	"	"
		"	66			IN	2.7 V	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	67			2.7 V	IN	2.7 V	OUT	"							"	2B to 2Y	"	"	"
		"	68			2.7 V	2.7 V	IN	OUT	"							"	2C to 2Y	"	"	"
		"	69							"	OUT	IN	2.7 V	2.7 V			"	3A to 3Y	"	"	"
		"	70							"	OUT	2.7 V	IN	2.7 V			"	3B to 3Y	"	"	"
		"	71							"	OUT	2.7 V	2.7 V	IN			"	3C to 3Y	"	"	"
10 T_C = +125°C	t_{PLH} t_{PHL}	Same tests, and terminal conditions as subgroup 9 except T_C = +125°C.																	2 3	12 11	" "
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																				

1/ Pins not referenced are N/C.

2/ I_{IL} limits in µA as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-100	0/-100	0/-100

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

4/ I_O min/max limit for circuits B and C shall be: -30/-110 mA.

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TABLE III. Group A inspection for device type 07.

Terminal conditions (pins not designated may be high > 2.0 V or low < 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2 1/ Test no.	1 2 IA	2 3 IB	3 1/ NC	4 6 IC	5 8 ID	6 9 IY	7 10 GND	8 12 2Y	9 13 2A	10 14 2B	11 1/ NC	12 18 2C	13 19 2D	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	2.0 V		2.0 V	2.0 V	-1.0 mA	GND							4.5 V	1Y	2.4		V dc
	"	"	2	2.0 V	0.8 V		2.0 V	2.0 V	-1.0 mA	"							"	1Y	"		"
	"	"	3	2.0 V	2.0 V		0.8 V	2.0 V	-1.0 mA	"							"	1Y	"		"
	"	"	4	2.0 V	2.0 V		2.0 V	0.8 V	-1.0 mA	"							"	1Y	"		"
	"	"	5							"	-1.0 mA	0.8 V	2.0 V		2.0 V	2.0 V	"	2Y	"		"
	"	"	6							"	-1.0 mA	2.0 V	0.8 V		2.0 V	2.0 V	"	2Y	"		"
	"	"	7							"	-1.0 mA	2.0 V	2.0 V		0.8 V	2.0 V	"	2Y	"		"
	"	"	8							"	-1.0 mA	2.0 V	2.0 V		2.0 V	0.8 V	"	2Y	"		"
	V_{OL}	3007	9	2.0 V	2.0 V		2.0 V	2.0 V	12 mA	"							"	1Y		0.4	"
		3007	10							"	12 mA	2.0 V	2.0 V		2.0 V	2.0 V	"	2Y		0.4	"
	V_{IC}		11	-18 mA						"							"	1A		1.5	"
			12		-18 mA					"							"	1B		"	"
			13				-18 mA			"							"	1C		"	"
			14					-18 mA		"							"	1D		"	"
			15							"		-18 mA					"	2A		"	"
			16							"			-18 mA				"	2B		"	"
			17							"					-18 mA		"	2C		"	"
			18							"						-18 mA	"	2D		"	"
	I_{IL}	3009	19	0.4 V						"							5.5 V	1A	2/	2/	µA
	"	"	20		0.4 V					"							"	1B	"	"	"
	"	"	21				0.4 V			"							"	1C	"	"	"
	"	"	22					0.4 V		"							"	1D	"	"	"
	"	"	23							"		0.4 V					"	2A	"	"	"
	"	"	24							"			0.4 V				"	2B	"	"	"
	"	"	25							"					0.4 V		"	2C	"	"	"
	"	"	26							"						0.4 V	"	2D	"	"	"
	I_{IH1}	3010	27	2.7 V						"							"	1A		20	"
	"	"	28		2.7 V					"							"	1B		"	"
	"	"	29				2.7 V			"							"	1C		"	"
	"	"	30					2.7 V		"							"	1D		"	"
	"	"	31							"		2.7 V					"	2A		"	"
	"	"	32							"			2.7 V				"	2B		"	"
	"	"	33							"					2.7 V		"	2C		"	"
	"	"	34							"						2.7 V	"	2D		"	"
	I_{IH2}	"	35	7.0 V						"							"	1A		100	"
	"	"	36		7.0 V					"							"	1B		"	"
	"	"	37				7.0 V			"							"	1C		"	"
	"	"	38					7.0 V		"							"	1D		"	"
	"	"	39							"		7.0 V					"	2A		"	"
	"	"	40							"			7.0 V				"	2B		"	"
	"	"	41							"					7.0 V		"	2C		"	"
	"	"	42							"						7.0 V	"	2D		"	"
	I_O	3011 3/ 4/	43	GND	GND		GND	GND	2.25 V	"							"	1Y	-30	-112	mA
			44							"	2.25 V	GND	GND		GND	GND	"	2Y	-30	-112	"
	I_{CCH}	3005	45	GND	GND		GND	GND		"		GND	GND		GND	GND	"	V_{CC}		0.8	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 07 - Continued.

Terminal conditions (pins not designated may be high > 2.0 V or low < 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	1/	6	8	9	10	12	13	14	1/	18	19	20		Min	Max	
			Test no.	1A	1B	NC	1C	1D	1Y	GND	2Y	2A	2B	NC	2C	2D	V_{CC}				
1 T_C = +25°C	I_{CCL}	3005	46	4.5 V	4.5 V		4.5 V	4.5 V		GND		4.5 V	4.5 V		4.5 V	4.5 V	5.5 V	V_{CC}		3.9	mA
2	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PLH}	3003 Fig. 4	47	IN	2.7 V		2.7 V	2.7 V	OUT	GND							5.0 V	1A to 1Y	2	8	ns
		"	48	2.7 V	IN		2.7 V	2.7 V	OUT	"							"	1B to 1Y	"	"	"
		"	49	2.7 V	2.7 V		IN	2.7 V	OUT	"							"	1C to 1Y	"	"	"
		"	50	2.7 V	2.7 V		2.7 V	IN	OUT	"							"	1D to 1Y	"	"	"
		"	51							"	OUT	IN	2.7 V		2.7 V	2.7 V	"	2A to 2Y	"	"	"
		"	52							"	OUT	2.7 V	IN		2.7 V	2.7 V	"	2B to 2Y	"	"	"
		"	53							"	OUT	2.7 V	2.7 V		IN	2.7 V	"	2C to 2Y	"	"	"
		"	54							"	OUT	2.7 V	2.7 V		2.7 V	IN	"	2D to 2Y	"	"	"
	t_{PHL}	"	55	IN	2.7 V		2.7 V	2.7 V	OUT	"							"	1A to 1Y	3	"	"
		"	56	2.7 V	IN		2.7 V	2.7 V	OUT	"							"	1B to 1Y	"	"	"
		"	57	2.7 V	2.7 V		IN	2.7 V	OUT	"							"	1C to 1Y	"	"	"
		"	58	2.7 V	2.7 V		2.7 V	IN	OUT	"							"	1D to 1Y	"	"	"
		"	59							"	OUT	IN	2.7 V		2.7 V	2.7 V	"	2A to 2Y	"	"	"
		"	60							"	OUT	2.7 V	IN		2.7 V	2.7 V	"	2B to 2Y	"	"	"
		"	61							"	OUT	2.7 V	2.7 V		IN	2.7 V	"	2C to 2Y	"	"	"
		"	62							"	OUT	2.7 V	2.7 V		2.7 V	IN	"	2D to 2Y	"	"	"
10 T_C = +125°C	t_{PLH} t_{PHL}	Same tests and terminal conditions as subgroup 9 except T_C = +125°C.																	2 3	10 10	" "
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																				

1/ Pins not referenced are N/C.

2/ I_{IL} limits in μA as follows:

Test	Circuit		
	A	B	C
I_{IL}	0/-100	0/-100	0/-100

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

4/ I_O min/max limit for circuits B and C shall be: -30/-110 mA.

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TABLE III. Group A inspection for device type 08.

Terminal conditions (pins not designated may be high > 2.0 V or low < 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ ? Test no.	1 1A	2 1B	3 1Y	4 2A	5 2B	6 2Y	7 GND	8 3Y	9 3A	10 3B	11 4Y	12 4A	13 4B	14 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	0.8 V	-1.0 mA				GND							4.5 V	1Y	2.4		V dc
		"	2	0.8 V	2.0 V	-1.0 mA				"							"	1Y	"		"
		"	3				2.0 V	0.8 V	-1.0 mA	"							"	2Y	"		"
		"	4				0.8 V	2.0 V	-1.0 mA	"							"	2Y	"		"
		"	5							"	-1.0 mA	2.0 V	0.8 V				"	3Y	"		"
		"	6							"	-1.0 mA	0.8 V	2.0 V				"	3Y	"		"
		"	7							"				-1.0 mA	2.0 V	0.8 V	"	4Y	"		"
		"	8							"				-1.0 mA	0.8 V	2.0 V	"	4Y	"		"
	V_{OL}	3007	9	0.8 V	0.8 V	12 mA				"							"	1Y		0.4	"
		"	10				0.8 V	0.8 V	12 mA	"							"	2Y		"	"
		"	11							"	12 mA	0.8 V	0.8 V				"	3Y		"	"
		"	12							"				12 mA	0.8 V	0.8 V	"	4Y		"	"
	V_{IC}		13	-18 mA						"							"	1A		-1.5	"
			14		-18 mA					"							"	1B		"	"
			15				-18 mA			"							"	2A		"	"
			16					-18 mA		"							"	2B		"	"
			17							"		-18 mA					"	3A		"	"
			18							"			-18 mA				"	3B		"	"
			19							"					-18 mA		"	4A		"	"
			20							"						-18 mA	"	4B		"	"
	I_{IL}	3009	21	0.4 V						"							5.5 V	1A	2/	2/	uA
		"	22		0.4 V					"							"	1B	"	"	"
		"	23				0.4 V			"							"	2A	"	"	"
		"	24					0.4 V		"							"	2B	"	"	"
		"	25							"		0.4 V					"	3A	"	"	"
		"	26							"			0.4 V				"	3B	"	"	"
		"	27							"					0.4 V		"	4A	"	"	"
		"	28							"						0.4 V	"	4B	"	"	"
	I_{IH1}	3010	29	2.7 V						"							"	1A		20	"
		"	30		2.7 V					"							"	1B		"	"
		"	31				2.7 V			"							"	2A		"	"
		"	32					2.7 V		"							"	2B		"	"
		"	33							"		2.7 V					"	3A		"	"
		"	34							"			2.7 V				"	3B		"	"
		"	35							"					2.7 V		"	4A		"	"
		"	36							"						2.7 V	"	4B		"	"
	I_{IH2}	"	37	7.0 V						"							"	1A		100	"
		"	38		7.0 V					"							"	1B		"	"
		"	39				7.0 V			"							"	2A		"	"
		"	40					7.0 V		"							"	2B		"	"
		"	41							"		7.0 V					"	3A		"	"
		"	42							"			7.0 V				"	3B		"	"
		"	43							"					7.0 V		"	4A		"	"
		"	44							"						7.0 V	"	4B		"	"
	I_O	3011 3/ 4/	45	4.5 V	4.5 V	2.25 V				"							"	1Y	-30	-112	mA
		"	46				4.5 V	4.5 V	2.25 V	"							"	2Y	"	"	"
		"	47							"	2.25 V	4.5 V	4.5 V				"	3Y	"	"	"
		"	48							"				2.25 V	4.5 V	4.5 V	"	4Y	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 08 - Continued.

Terminal conditions (pins not designated may be high > 2.0 V or low < 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 1A	2 2 1B	3 3 1Y	4 6 2A	5 8 2B	6 9 2Y	7 10 GND	8 12 3Y	9 13 3A	10 14 3B	11 16 4Y	12 18 4A	13 19 4B	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{CCH}	3005	49	4.5 V	4.5 V		4.5 V	4.5 V		GND		4.5 V	4.5 V		4.5 V	4.5 V	5.5 V	V_{CC}		5.0	mA
	I_{CCL}	3005	50	GND	GND		GND	GND		GND		GND	GND		GND	GND	5.5 V	V_{CC}		10.6	mA
2	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PLH}	3003 Fig. 4	51	IN	GND	OUT				GND							5.0 V	1A to 1Y	2	10	ns
		"	52	GND	IN	OUT				"							"	1B to 1Y	"	"	"
		"	53				IN	GND	OUT	"							"	2A to 2Y	"	"	"
		"	54				GND	IN	OUT	"							"	2B to 2Y	"	"	"
		"	55							"	OUT	IN	GND				"	3A to 3Y	"	"	"
		"	56							"	OUT	GND	IN				"	3B to 3Y	"	"	"
		"	57							"				OUT	IN	GND	"	4A to 4Y	"	"	"
		"	58							"				OUT	GND	IN	"	4B to 4Y	"	"	"
	t_{PHL}	"	59	IN	GND	OUT				"							"	1A to 1Y	"	13	"
		"	60	GND	IN	OUT				"							"	1B to 1Y	"	"	"
		"	61				IN	GND	OUT	"							"	2A to 2Y	"	"	"
		"	62				GND	IN	OUT	"							"	2B to 2Y	"	"	"
		"	63							"	OUT	IN	GND				"	3A to 3Y	"	"	"
		"	64							"	OUT	GND	IN				"	3B to 3Y	"	"	"
		"	65							"				OUT	IN	GND	"	4A to 4Y	"	"	"
		"	66							"				OUT	GND	IN	"	4B to 4Y	"	"	"
10 T_C = +125°C	t_{PLH} t_{PHL}	Same tests and terminal conditions as subgroup 9 except T_C = +125°C.																	2 2	12 15	" "
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																				

1/ Pins not referenced are N/C.

2/ I_{IL} limits in μA as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-100	0/-100	0/-100

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

4/ I_O min/max limit for circuits B and C shall be: -30/-110 mA.

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TABLE III. Group A inspection for device type 09, 10.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2 Test no.	Type 09	Type 10	1 2 1A	2 3 1Y	3 4 2A	4 6 2Y	5 8 3A	6 9 3Y	7 10 GND	8 12 4Y	9 13 4A	10 14 5Y	11 16 5A	12 18 6Y	13 19 6A	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH} 2/	3007	1	1		0.8 V	-1 mA	0.8 V	-1 mA	0.8 V	-1 mA	GND	-1 mA	0.8 V	-1 mA	0.8 V	-1 mA	0.8 V	4.5 V	1Y	2.4		V dc
		"	2									"							"	2Y	"		"
		"	3									"							"	3Y	"		"
		"	4									"							"	4Y	"		"
		"	5									"							"	5Y	"		"
		"	6									"							"	6Y	"		"
	V_{OL}	3006	7	1		2.0 V	12 mA	2.0 V	12 mA	2.0 V	12 mA	"	12 mA	2.0 V	12 mA	2.0 V	12 mA	2.0 V	"	1Y		0.4	V dc
		"	8	2								"							"	2Y		"	"
		"	9	3								"							"	3Y		"	"
		"	10	4								"							"	4Y		"	"
		"	11	5								"							"	5Y		"	"
		"	12	6								"							"	6Y		"	"
	V_{IC}		13	7		-18 mA		-18 mA		-18 mA		"		-18 mA		-18 mA		-18 mA	"	1A		-1.5	"
			14	8								"							"	2A		"	"
			15	9								"							"	3A		"	"
			16	10								"							"	4A		"	"
			17	11								"							"	5A		"	"
			18	12								"							"	6A		"	"
	I_{IL}	3009	19	13		0.4 V		0.4 V		0.4 V		"		0.4 V		0.4 V		0.4 V	5.5 V	1A	3/	3/	µA
		"	20	14								"							"	2A	"	"	"
		"	21	15								"							"	3A	"	"	"
		"	22	16								"							"	4A	"	"	"
		"	23	17								"							"	5A	"	"	"
		"	24	18								"							"	6A	"	"	"
	I_{IH1}	3010	25	19		2.7 V		2.7 V		2.7 V		"		2.7 V		2.7 V		2.7 V	"	1A		20	"
		"	26	20								"							"	2A		"	"
		"	27	21								"							"	3A		"	"
		"	28	22								"							"	4A		"	"
		"	29	23								"							"	5A		"	"
		"	30	24								"							"	6A		"	"
	I_{IH2}	"	31	25		7.0 V		7.0 V		7.0 V		"		7.0 V		7.0 V		7.0 V	"	1A		100	"
		"	32	26								"							"	2A		"	"
		"	33	27								"							"	3A		"	"
		"	34	28								"							"	4A		"	"
		"	35	29								"							"	5A		"	"
		"	36	30								"							"	6A		"	"
	I_O 2/ 4/ 5/	3011	37			GND	2.25 V	GND	2.25 V	GND	2.25 V	"	2.25 V	GND	2.25 V	GND	2.25 V	GND	"	1Y	-30	-112	mA
		"	38									"							"	2Y	"	"	"
		"	39									"							"	3Y	"	"	"
		"	40									"							"	4Y	"	"	"
		"	41									"							"	5Y	"	"	"
		"	42									"							"	6Y	"	"	"
	I_{CEX} 6/	"		31		GND	5.5 V	GND	5.5 V	GND	5.5 V	"	5.5 V	GND	5.5 V	GND	5.5 V	GND	"	1Y		100	µA
		"		32								"							"	2Y		"	"
		"		33								"							"	3Y		"	"
		"		34								"							"	4Y		"	"
		"		35								"							"	5Y		"	"
		"		36								"							"	6Y		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 09, 10 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.		1 1A	2 1Y	3 2A	4 2Y	5 3A	6 3Y	7 GND	8 4Y	9 4A	10 5Y	11 5A	12 6Y	13 6A	14 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
			2	3	4	6	8	9	10	12	13	14	16	18	19	20						
			Type 09	Type 10																		
1 T_C = +25°C	I_{CCH}	3005	43	37	GND		GND		GND		GND		GND		GND		GND	5.5 V	V_{CC}		3	mA
	I_{CCL}	3005	44	38	4.5 V		4.5 V		4.5 V		GND		4.5 V		4.5 V		4.5 V	5.5 V	V_{CC}		12	mA
2	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																					
3	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																					
9 T_C = +25°C	t_{PLH}	3003 Fig. 4	45	39	IN	OUT	IN	OUT	IN	OUT	GND	OUT	IN	OUT	IN	OUT	IN	5.0 V	1A to 1Y	7/	7/	ns
		"	46	40							"							"	2A to 2Y	"	"	"
		"	47	41							"							"	3A to 3Y	"	"	"
		"	48	42							"							"	4A to 4Y	"	"	"
		"	49	43							"							"	5A to 5Y	"	"	"
		"	50	44							"							"	6A to 6Y	"	"	"
	t_{PHL}	"	51	45	IN	OUT	IN	OUT	IN	OUT	"	OUT	IN	OUT	IN	OUT	IN	"	1A to 1Y	8/	8/	"
		"	52	46							"							"	2A to 2Y	"	"	"
		"	53	47							"							"	3A to 3Y	"	"	"
		"	54	48							"							"	4A to 4Y	"	"	"
		"	55	49							"							"	5A to 5Y	"	"	"
		"	56	50							"							"	6A to 6Y	"	"	"
10 T_C = +125°C	t_{PLH} t_{PHL}	Same tests and terminal conditions as subgroup 9 except T_C = 125°C.																		9/ 10/	9/ 10/	"
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																					

1/ Pins not referenced are N/C.
2/ 09 only.
3/ I_{IL} limits in μA as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-100	0/-100	0/-100

4/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.
5/ I_O min/max limit for circuits B and C shall be: -30/-110 mA.
6/ 10 only.

		Min	Max	Unit
7/	for device type 09:	1	7	ns
	for device type 10:	5	30	ns
8/	for device type 09:	1	6	ns
	for device type 10:	2	10	ns
9/	for device type 09:	1	9	ns
	for device type 10:	5	35	ns
10/	for device type 09:	1	8	ns
	for device type 10:	2	12	ns

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TABLE III. Group A inspection for device type 11, 12.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no. Type 11	Type 12	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Unit
					2	3	4	8	9	10	12	13	14	16	18	19	20					
					1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	V_{CC}				
I T_C = +25°C	V_{OH} 2/	3007	1		2.0 V	-1 mA					GND							4.5 V	1Y	2.4		V dc
	"	"	2				2.0 V	-1 mA			"							"	2Y	"		"
	"	"	3						2.0 V	-1 mA	"							"	3Y	"		"
	"	"	4								"	-1 mA	2.0 V					"	4Y	"		"
	"	"	5								"			-1 mA	2.0 V			"	5Y	"		"
	"	"	6								"					-1 mA	2.0 V	"	6Y	"		"
	V_{OL}	3006	7	1	0.8 V	12 mA					"							"	1Y		0.4	"
	"	"	8	2			0.8 V	12 mA			"							"	2Y		"	"
	"	"	9	3					0.8 V	12 mA	"							"	3Y		"	"
	"	"	10	4							"	12 mA	0.8 V					"	4Y		"	"
	"	"	11	5							"			12 mA	0.8 V			"	5Y		"	"
	"	"	12	6							"					12 mA	0.8 V	"	6Y		"	"
	V_{IC}		13	7	-18 mA						"							"	1A		-1.5	"
	"		14	8			-18 mA				"							"	2A		"	"
	"		15	9					-18 mA		"							"	3A		"	"
	"		16	10							"		-18 mA					"	4A		"	"
	"		17	11							"				-18 mA			"	5A		"	"
	"		18	12							"						-18 mA	"	6A		"	"
	I_{IL}	3009	19	13	0.4 V						"							5.5 V	1A	3/	3/	mA
	"	"	20	14			0.4 V				"							"	2A	"	"	"
	"	"	21	15					0.4 V		"							"	3A	"	"	"
	"	"	22	16							"		0.4 V					"	4A	"	"	"
	"	"	23	17							"				0.4 V			"	5A	"	"	"
	"	"	24	18							"						0.4 V	"	6A	"	"	"
	I_{IH1}	3010	25	19	2.7 V						"							"	1A		20	"
	"	"	26	20			2.7 V				"							"	2A		"	"
	"	"	27	21					2.7 V		"							"	3A		"	"
	"	"	28	22							"		2.7 V					"	4A		"	"
	"	"	29	23							"				2.7 V			"	5A		"	"
	"	"	30	24							"						2.7 V	"	6A		"	"
	I_{IH2}	"	31	25	7.0 V						"							"	1A		100	"
	"	"	32	26			7.0 V				"							"	2A		"	"
	"	"	33	27					7.0 V		"							"	3A		"	"
	"	"	34	28							"		7.0 V					"	4A		"	"
	"	"	35	29							"				7.0 V			"	5A		"	"
	"	"	36	30							"						7.0 V	"	6A		"	"
	I_O 2/ 4/ 5/	3011	37		5.5 V	2.25 V					"							"	1Y	-30	-112	mA
	"	"	38				5.5 V	2.25 V			"							"	2Y	"	"	"
	"	"	39						5.5 V	2.25 V	"							"	3Y	"	"	"
	"	"	40								"	2.25 V	5.5 V					"	4Y	"	"	"
	"	"	41								"			2.25 V	5.5 V			"	5Y	"	"	"
	"	"	42								"					2.25 V	5.5 V	"	6Y	"	"	"
	I_{OH} 6/	"		31	5.0 V	5.5 V					"							"	1Y		100	µA
	"	"		32			5.0 V	5.5 V			"							"	2Y		"	"
	"	"		33					5.0 V	5.5 V	"							"	3Y		"	"
	"	"		34							"	5.5 V	5.0 V					"	4Y		"	"
	"	"		35							"			5.5 V	5.0 V			"	5Y		"	"
	"	"		36							"					5.5 V	5.0 V	"	6Y		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 11, 12 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2 Test no.	Type 11	Type 12	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Unit
						1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	V_{CC}				
1 T_C = +25°C	I_{CCH}	3005	43	37	4.5 V		4.5 V		4.5 V		GND		4.5 V		4.5 V		4.5 V	5.5 V	V_{CC}		6	mA
	I_{CCL}	3005	44	38	GND		GND		GND		GND		GND		GND		GND	5.5 V	V_{CC}		14	mA
2	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																						
9 T_C = +25°C	t_{PLH}	3003 Fig. 4	45	39	IN		OUT		IN		OUT		GND							5.0 V	1A to 1Y	7/	7/	ns
			46	40									"							"	2A to 2Y	"	"	"
			47	41								IN	"							"	3A to 3Y	"	"	"
			48	42									"	OUT	IN					"	4A to 4Y	"	"	"
			49	43									"			OUT	IN			"	5A to 5Y	"	"	"
			50	44									"					OUT	IN	"	6A to 6Y	"	"	"
	t_{PHL}	"	51	45	IN		OUT		IN		OUT		"							"	1A to 1Y	8/	8/	"
		"	52	46									"							"	2A to 2Y	"	"	"
		"	53	47						IN			"							"	3A to 3Y	"	"	"
		"	54	48									"	OUT	IN					"	4A to 4Y	"	"	"
		"	55	49									"			OUT	IN			"	5A to 5Y	"	"	"
		"	56	50									"					OUT	IN	"	6A to 6Y	"	"	"
10 T_C = +125°C	t_{PLH} t_{PHL}	Same tests and terminal conditions as subgroup 9 except T_C = +125°C.																				9/ 10/	9/ 10/	"
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																						

1/ Pins not referenced are N/C.
2/ 11 only.
3/ I_{IL} limits in μA as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-100	0/-100	0/-100

4/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.
5/ I_O min/max limit for circuits B and C shall be: -30/-110 mA.
6/ 12 only.

		Min	Max	Unit
7/	for device type 11:	1	8	ns
	for device type 12:	5	30	ns
8/	for device type 11:	1	8	ns
	for device type 12:	2	12	ns
9/	for device type 11:	1	10	ns
	for device type 12:	5	35	ns
10/	for device type 11:	1	10	ns
	for device type 12:	2	14	ns

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
V_{IN}	Voltage level at an input terminal.
V_{IC}	Input clamp voltage.
I_{IN}	Current flowing into an input terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

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Military device type	Generic-industry type
01	54ALS1000/54ALS37
02	54ALS1002/54ALS28
03	54ALS1003/54ALS38
04	54ALS1008
05	54ALS1010
06	54ALS1011
07	54ALS1020/54ALS40
08	54ALS1032
09	54ALS1004
10	54ALS1005
11	54ALS1034
12	54ALS1035

6.7 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit		
	A	B	C
	Texas Instruments	Motorola	National
01	X		
02	X		
03	X		
04	X		
05	X		
06	X		
07	X		
08	X		
09	X		
10	X		
11	X		
12	X		

Custodians:
Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-F636)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
M38510/384(USAF)	Microcircuits, Digital, Advanced Low Power Schottky TTL, AND, OR, NAND, AND NOR Gates Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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